SA-AEF: Source-Aware Acoustic Enhanced Fusion for Cross-Regional Wu Dialect Vowel Recognition

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Abstract. Cross-regional Wu dialect vowel recognition, as a fine-grained speech classification task, aims to identify vowel categories under regional acoustic variation. Existing methods often suffer from inadequate exploitation of heterogeneous acoustic cues, weak adaptation to sample-dependent cue reliability, and the scarcity of fine-grained cross-regional Wu dialect vowel datasets. To address these challenges, we propose a source-aware acoustic enhanced fusion framework for Wu dialect vowel recognition, namely SA-AEF. First, in the Context-Aware Gating (CAG) module, we construct prompt-enhanced source-aware tokens and adaptively recalibrate them according to the acoustic context, enabling the model to preserve source identities and emphasize reliable acoustic information. Second, in the Acoustic Enhanced Fusion (AEF) module, cross-source interactions among the recalibrated tokens are modeled to produce a discriminative fused representation for vowel classification. In addition, to alleviate the scarcity of fine-grained cross-regional Wu dialect vowel datasets, we construct WuVowelSeg, a fine-grained vowel benchmark covering 45 survey locations across 14 regions. Extensive experiments are conducted on WuVowelSeg under cross-region testing. The results demonstrate that SA-AEF achieves superior performance among the compared methods, obtaining 74.3% accuracy, 64.4% Macro-F1, and 74.2% Weighted-F1. The code is publicly available at `https://github.com/scilabret/SA-AEF`.

Keywords: Vowel Recognition · Acoustic Feature Fusion · Source-Aware Fusion · WuVowelSeg

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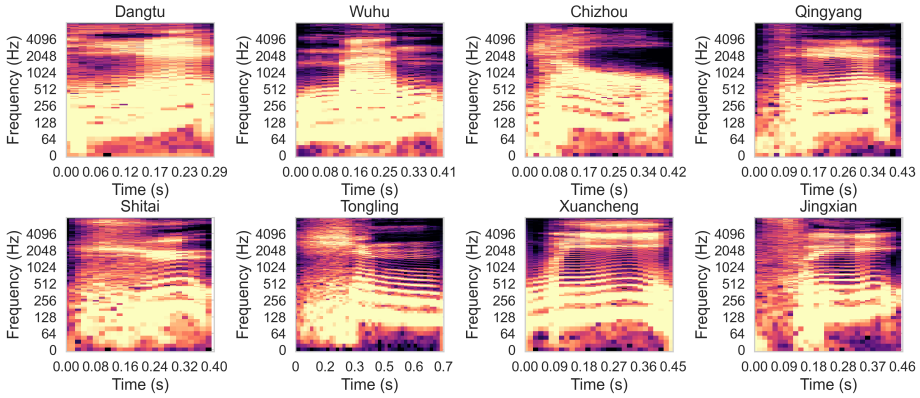


Fig. 1. Spectrogram examples of the same IPA-normalized vowel category from eight representative Wu dialect regions, showing regional acoustic differences in vowel realizations.

1 Introduction

Vowel recognition aims to identify fine-grained vowel categories from speech signals and plays an important role in phonetic analysis, dialect research, and speech technology. Classical phonetic studies show that vowel categories are closely related to formant patterns and spectral characteristics [1], while these acoustic cues may vary substantially across speakers and regions [2]. This variation makes cross-regional vowel recognition challenging, especially for Wu dialects, whose vowel systems exhibit rich regional diversity [3, 4]. As illustrated in Fig. 1, vowel segments from representative Wu dialect regions show distinct spectral patterns, implying that robust vowel recognition for Wu dialects demands effective handling of regional acoustic variation and fine-grained phonetic ambiguity.

Existing approaches adopt hand-crafted acoustic descriptors and neural speech representations for vowel analysis. MFCCs, formants and spectral statistics deliver interpretable low-level acoustic features [5–7], while pre-trained models including wav2vec 2.0 [8], HuBERT [9], WavLM [10] and Whisper [11] yield powerful transferable speech embeddings. These heterogeneous representations are generally utilized separately or combined through feature-level fusion strategies. Static concatenation and pooling provide fixed feature combinations, while attention-based fusion models source-level importance. This motivates a source-aware fusion scheme that retains acoustic-source identity, adaptively recalibrates acoustic features, and captures cross-source interactions.

To tackle the above issues, this paper proposes **SA-AEF**, a Source-Aware Acoustic Enhanced Fusion framework tailored for cross-regional Wu dialect vowel recognition. Multiple heterogeneous acoustic features are converted into source-aware tokens via dedicated projection layers and learnable source prompts. On the basis of prompt-enhanced tokens, the **Context-Aware Gating (CAG)** module conducts sample-dependent and dimension-wise recalibration guided by

global acoustic context, and the **Acoustic Enhanced Fusion (AEF)** module models cross-source interactions among calibrated token sequences. Together, these components organize source identity preservation, acoustic-context-based recalibration, and cross-source interaction within a unified fusion process tailored to regional acoustic variation. For standardized experimental evaluation, we build **WuVowelSeg**, a vowel-level Wu dialect dataset collected from 45 survey locations across 14 regions, annotated with standard IPA labels and featuring natural regional data imbalance. Experimental results on WuVowelSeg demonstrate that SA-AEF achieves superior performance among the compared conventional machine learning classifiers, pre-trained speech models, and fusion methods under cross-region testing.

The main contributions of this work are summarized as follows:

- We propose SA-AEF, a source-aware fusion framework that adaptively integrates heterogeneous acoustic representations for cross-regional Wu dialect vowel recognition.
- We design CAG to recalibrate source-aware tokens and AEF to model cross-source interactions for robust vowel representation.
- We construct WuVowelSeg, a fine-grained Wu dialect vowel recognition dataset covering 45 survey locations across 14 regions, and provide systematic cross-region evaluation.

2 Related Work

2.1 Vowel Recognition and Acoustic Representations

Vowel recognition has been widely applied to phonetic analysis, pronunciation assessment, and speech recognition. Traditional methods combine hand-crafted features such as formants, MFCCs, pitch, energy, duration, and spectral statistics [1, 5, 7] with SVM, HMM, k-NN, or neural classifiers [12]. Deep acoustic models learn frame-level alignments and contextual representations using CTC, Transformer, and Conformer architectures [13–15]. Pre-trained speech models, including wav2vec 2.0, HuBERT, WavLM, and Whisper, provide transferable representations for downstream speech tasks [8–11]. Multi-source acoustic representations are commonly integrated through feature concatenation, channel or dimension recalibration, and attention-based fusion. These strategies model complementary information across acoustic features, while acoustic-source identity preservation and sample-dependent dimension-wise recalibration remain insufficiently investigated in cross-regional vowel recognition.

2.2 Cross-Domain Dialect Recognition

Cross-domain dialect recognition improves robustness when training and test data come from different regions, speakers, or dialectal varieties. Domain adaptation, transfer learning, region-aware modeling, and multi-dialect training have

been used to reduce acoustic mismatch [16–18]. For Wu dialect, acoustic studies of Hangzhou, Rui'an, and Taizhou show complex region-specific vowel patterns [19–21]. Yue et al. constructed a multi-regional Wu dialect corpus and introduced two-level phoneme mapping, regional label embedding, and transfer learning for Wu dialect speech recognition [22]. These studies support dialect speech modeling, but fine-grained vowel recognition under cross-regional acoustic shifts remains less explored.

3 Method

This section presents **SA-AEF**, a Source-Aware Acoustic Enhanced Fusion framework for cross-regional vowel recognition. As shown in Fig. 2, in the proposed SA-AEF framework, multiple acoustic sources are organized into source-aware acoustic tokens through source-specific projections and learnable source prompts. The prompt-enhanced tokens are then adaptively recalibrated via the Context-Aware Gating (CAG) module, and cross-source interactions are further modeled through the Acoustic Enhanced Fusion (AEF) module for vowel classification.

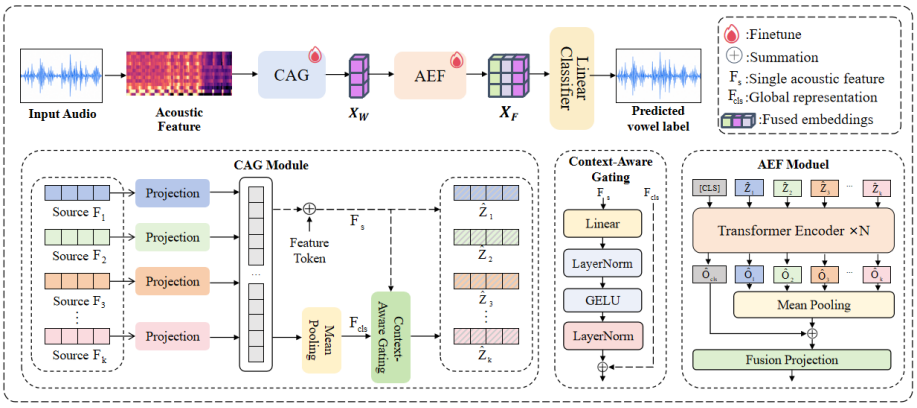


Fig. 2. Overall architecture of the proposed SA-AEF framework. Multi-source acoustic features are projected into a shared token space and enhanced with learnable source prompts. CAG recalibrates the source-aware tokens according to the acoustic context, and AEF models cross-source interactions for vowel classification.

3.1 Source-Aware Tokenization and Context-Aware Gating

Given a vowel segment x, SA-AEF extracts a set of heterogeneous acoustic sources $\{\mathbf{F}_j\}_{j=1}^{n}$, where $\mathbf{F}_j \in \mathbb{R}^{d_j}$ denotes the feature vector of the j-th acoustic source, d_j is its original feature dimension, and n is the number of acoustic

sources. These sources describe the same vowel segment from different acoustic perspectives, but may differ in dimension, distribution, and robustness to regional pronunciation variation. Instead of directly concatenating them, SA-AEF maps each source into a shared d-dimensional token space.

For the j-th acoustic source, a source-specific projection layer is used and a learnable source prompt is injected:

$$\mathbf{S}_j = \mathrm{Proj}_j(\mathbf{F}_j) + \mathbf{P}_j, \quad j = 1, 2, \ldots, n, \tag{1}$$

where $\mathrm{Proj}_j(\cdot)$ denotes the projection layer for the j-th source, $\mathbf{P}_j \in \mathbb{R}^d$ is its learnable source prompt, and $\mathbf{S}_j \in \mathbb{R}^d$ is the resulting source-aware token. Here, "source" refers to acoustic feature sources rather than geographic regions. The prompt acts as an identity embedding, allowing tokens in the shared latent space to preserve their source-specific characteristics and providing a source reference for subsequent acoustic-context-based recalibration.

In cross-region vowel recognition, the reliability of each acoustic source varies across vowel segments and regional pronunciation patterns. Therefore, CAG recalibrates the prompt-enhanced source tokens according to the acoustic context of the current input. The global acoustic context is first summarized by averaging all source-aware tokens:

$$\mathbf{c} = \frac{1}{n} \sum_{j=1}^{n} \mathbf{S}_j, \tag{2}$$

where $\mathbf{c} \in \mathbb{R}^d$ denotes the global context vector.

For each source token, CAG estimates a dimension-wise gating vector by jointly considering the source-specific token and the global context:

$$\mathbf{G}_j = \sigma\left(\mathrm{MLP}_g\left([\mathbf{S}_j; \mathbf{c}]\right)\right), \tag{3}$$

where $[\cdot;\cdot]$ denotes concatenation, $\sigma(\cdot)$ is the sigmoid function, $\mathrm{MLP}_g(\cdot)$ is a lightweight gating network, and $\mathbf{G}_j \in (0,1)^d$ is the dimension-wise gate for the j-th source. The shared context summarizes the current combination of acoustic cues, while each gate remains associated with its corresponding source token.

The gate is then converted into a residual modulation signal to obtain the recalibrated source token:

$$\boldsymbol{\Delta}_j = 2\mathbf{G}_j - 1, \quad \hat{\mathbf{Z}}_j = \mathbf{S}_j + \boldsymbol{\Delta}_j \odot \mathbf{S}_j, \tag{4}$$

where $\boldsymbol{\Delta}_j \in (-1,1)^d$ denotes the residual modulation signal, $\odot$ denotes element-wise multiplication, and $\hat{\mathbf{Z}}_j \in \mathbb{R}^d$ is the recalibrated source token. In this way, CAG can enhance or suppress different dimensions of each source token according to the current acoustic context. When $\mathbf{G}_j = 0.5$, the modulation becomes zero and the original source-aware token is preserved.

3.2 Acoustic Enhanced Fusion

After CAG, SA-AEF obtains a sequence of recalibrated source tokens $\{\hat{\mathbf{Z}}_j\}_{j=1}^{n}$. Although each token has been adjusted according to the current acoustic context,

the acoustic sources still need to interact with each other to form a unified vowel representation. Therefore, AEF treats the recalibrated tokens as a compact source-token sequence and prepends a learnable global fusion token:

$$\mathbf{T}^0 = [\mathbf{T}^0_{\text{cls}}; \hat{\mathbf{Z}}_1; \hat{\mathbf{Z}}_2; \ldots; \hat{\mathbf{Z}}_n], \tag{5}$$

where $\mathbf{T}^0$ denotes the initial source-token sequence, and $\mathbf{T}^0_{\text{cls}}$ is used to aggregate global information from all acoustic sources.

The token sequence is processed by L Transformer encoder layers:

$$\mathbf{T}^\ell = \text{TransformerEncoder}^\ell(\mathbf{T}^{\ell-1}), \quad \ell = 1, 2, \ldots, L, \tag{6}$$

where $\mathbf{T}^\ell$ denotes the output sequence of the ℓ-th Transformer encoder layer, and L is the total number of layers. The output sequence is denoted as:

$$\mathbf{T}^L = [\mathbf{O}_{\text{cls}}; \mathbf{O}_1; \mathbf{O}_2; \ldots; \mathbf{O}_n], \tag{7}$$

where $\mathbf{T}^L$ is the final output token sequence, $\mathbf{O}_{\text{cls}}$ is the output of the global fusion token, and $\mathbf{O}_j$ corresponds to the j-th acoustic source token. Self-attention is performed over the compact sequence of source-level tokens, enabling AEF to model dependencies among source-aware and context-recalibrated acoustic representations.

To obtain the final fused representation, we combine the global token output with the mean-pooled source outputs:

$$\mathbf{X}_F = \text{MLP}\left([\mathbf{O}_{\text{cls}}; \frac{1}{n}\sum_{j=1}^{n}\mathbf{O}_j]\right). \tag{8}$$

The global token captures cross-source dependencies, while the mean-pooled source representation preserves the overall acoustic information. Their combination provides a compact and discriminative fused representation $\mathbf{X}_F$ for vowel classification.

3.3 Training and Inference

The fused representation $\mathbf{X}_F$ is fed into a lightweight classifier to predict the vowel category:

$$\mathbf{s} = \mathbf{W}_c\mathbf{X}_F + \mathbf{b}_c, \tag{9}$$

where $\mathbf{s}$ denotes the logits over vowel categories. For a mini-batch of M samples, the model is optimized with the standard cross-entropy loss:

$$\mathcal{L}_{cls} = -\frac{1}{M}\sum_{i=1}^{M}\log\frac{\exp(s_{i,y_i})}{\sum_{k=1}^{C}\exp(s_{i,k})}. \tag{10}$$

Here, C is the number of vowel categories, y_i is the ground-truth category of the i-th sample, and $s_{i,k}$ denotes its logit for the k-th category. During training, the projection layers, source prompts, CAG, AEF, and classifier are jointly optimized, while all acoustic feature extractors are kept frozen. During inference, SA-AEF predicts vowel labels without requiring any regional label.

4 Experiments

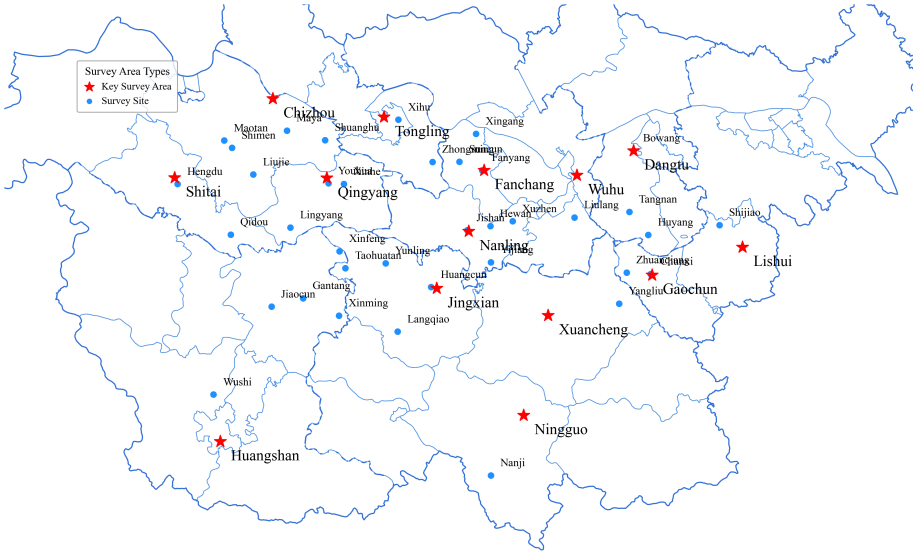


Fig. 3. Geographic distribution of WuVowelSeg regions and survey locations. Red stars denote key survey areas, and blue dots denote survey locations.

4.1 Dataset and Protocol

To address the lack of fine-grained resources for cross-regional Wu dialect vowel recognition, we construct **WuVowelSeg** from field-collected Wu dialect recordings in southern Anhui, Jiangsu, and Zhejiang, where Wu dialects exhibit rich regional vowel variation [3, 4]. As shown in Fig. 3, WuVowelSeg covers 45 survey locations grouped into 14 regions and includes recordings from 127 native speakers. All recordings are sampled at 16 kHz. The recordings were annotated in Praat by annotators with phonetic expertise. Vowel boundaries were determined from the waveforms and spectrograms based on periodicity, formant structure, and syllabic context. The annotations were subsequently checked by the research team, and disagreements were resolved through manual review. Syllable segmentation, vowel extraction, and IPA-based label normalization were then performed. Segments with missing boundaries, inconsistent IPA labels, severe background noise, invalid waveform–TextGrid alignment, or insufficient usable vowel content were removed during quality filtering. The resulting dataset contains 170,405 extracted vowel-level waveform segments from approximately 43,000 original audio clips, among which 149,199 segments are retained as valid TextGrid-aligned vowel segments.

Table 1. Regional data scale of WuVowelSeg. Sites, Spk., Seg., and Valid Seg. denote the numbers of survey locations, native speakers, extracted vowel-level waveform segments, and valid TextGrid-aligned vowel segments retained after quality filtering, respectively.

Region	Sites	Spk.	Seg.	Valid Seg.	Region	Sites	Spk.	Seg.	Valid Seg.
Dangtu	3	9	11932	9280	Wuhu	1	3	3977	3977
Chizhou	5	16	21205	17230	Qingyang	4	11	14583	11931
Shitai	2	13	16908	15582	Tongling	2	8	12977	7679
Xuancheng	1	3	3978	3978	Jingxian	5	15	19925	18600
Fanchang	3	7	9282	9282	Nanling	4	11	14583	14583
Huangshan	5	16	21212	21212	Ningguo	4	4	5304	2652
Gaochun	2	7	9237	7911	Lishui	4	4	5302	5302

Table 1 reports the regional data scale of WuVowelSeg. Each region consists of one or more survey locations, and the numbers of speakers and vowel segments vary across regions due to the practical constraints of field collection. This natural imbalance makes cross-region vowel recognition more challenging. All experiments are conducted on the valid TextGrid-aligned vowel segments. The models are trained and evaluated on fine-grained IPA-based vowel labels, while the four vowel-height groups, i.e., High, Mid-high, Mid-low, and Low, are used only for phonetic-level visualization and interpretability analysis. To evaluate cross-region generalization, each region is used in turn as the unseen target region, while samples from the remaining 13 regions are used for training. Final results are averaged over all 14 target regions.

4.2 Experimental Setup

For each vowel segment, SA-AEF uses three acoustic sources, corresponding to $n = 3$ in Sec. 3: 512-dimensional segment-level Whisper embeddings obtained by mean-pooling the frozen encoder outputs [11], 39-dimensional MFCC features [5], and 8-dimensional statistical acoustic descriptors [7]. All acoustic features are standardized using statistics computed only from the training regions before being fed into the fusion models. For traditional machine learning baselines, the concatenated 559-dimensional feature vector is standardized and reduced to 128 dimensions by PCA fitted only on the training regions. We compare with conventional classifiers, frozen pre-trained speech encoders, and fusion-based baselines.

For pre-trained speech baselines, frame-level encoder outputs are mean-pooled into segment-level representations and fed into the same classifier. For fusion baselines, Concat Fusion directly concatenates the three acoustic sources, while Attention Fusion learns source-level attention weights. In SA-AEF, all acoustic feature extractors are frozen, and only the projection layers, source prompts, CAG module, AEF module, and classifier are optimized with cross-entropy loss. The source-token dimension is set to 256, and AEF uses two Transformer encoder layers with four attention heads. All neural models are trained with AdamW for

Table 2. Average performance over 14 target regions on WuVowelSeg (%). The best and second-best results are highlighted in bold and underlined, respectively.

Method	Acc.	Macro-F1	Weighted-F1
Traditional acoustic-feature classifiers			
Logistic Regression [23]	69.1	62.1	70.3
Random Forest [24]	68.2	56.7	69.5
XGBoost [25]	69.8	59.7	71.8
LightGBM [26]	70.9	60.0	68.9
SVM [27]	71.9	61.4	72.1
Frozen pre-trained speech representations			
WavLM [10]	71.3	59.7	72.4
HuBERT [9]	<u>72.6</u>	62.7	73.0
wav2vec 2.0 [8]	70.8	58.9	67.9
Whisper [11]	71.9	60.2	<u>73.6</u>
Multi-source fusion methods			
Concat Fusion	71.3	62.5	72.6
Attention Fusion	72.3	<u>63.2</u>	73.3
SA-AEF (Ours)	**74.3**	**64.4**	**74.2**

100 epochs. Hyperparameters are selected using only the training regions, and target-region samples are not used for model selection. Accuracy, Macro-F1, and Weighted-F1 are computed on each target region and then averaged over all 14 target regions.

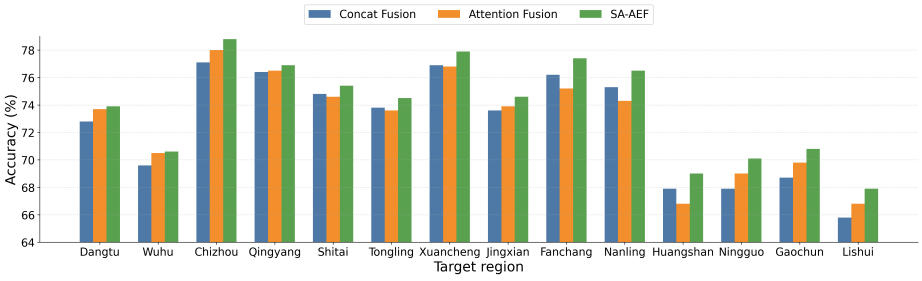


Fig. 4. Accuracy comparison of different fusion strategies across 14 target regions under cross-region evaluation. Each group corresponds to one target region, and the three bars denote Concat Fusion, Attention Fusion, and the proposed SA-AEF, respectively.

Table 3. Ablation and fusion analysis on the Dangtu target-region case study (%).

Method	Prompt	CAG	AEF	Acc.	Macro-F1	Weighted-F1
Whisper only	×	×	×	68.1	61.1	67.8
Whisper + MFCC	×	×	×	68.8	60.8	68.3
Whisper + MFCC + Stat.	×	×	×	71.8	62.5	71.5
SA-AEF w/o Prompt	×	✓	✓	72.3	61.6	72.0
SA-AEF w/o AEF	✓	✓	×	71.7	62.2	71.4
SA-AEF w/o CAG	✓	×	✓	72.7	62.1	72.2
SA-AEF	✓	✓	✓	**73.3**	**63.4**	**73.1**

4.3 Main Results

Table 2 compares SA-AEF with traditional acoustic-feature classifiers, frozen pre-trained speech representation baselines, and multi-source fusion methods under the leave-one-region-out protocol. SA-AEF achieves the best performance on all three metrics, obtaining 74.3% Acc., 64.4% Macro-F1, and 74.2% Weighted-F1. Compared with the strongest non-fusion baseline, SA-AEF improves Acc. by an absolute gain of 1.7% over HuBERT. Compared with Attention Fusion, SA-AEF further improves Acc., Macro-F1, and Weighted-F1 by absolute gains of 2.0%, 1.2%, and 0.9%, respectively.

These results indicate that hand-crafted features and frozen pre-trained representations both provide useful cues for Wu dialect vowel recognition. Concat Fusion and Attention Fusion also benefit from heterogeneous acoustic sources, while SA-AEF further integrates acoustic-source identity preservation, adaptive source recalibration, and cross-source interaction modeling.

4.4 Fusion and Ablation Analysis

We further analyze the fusion mechanism and component contributions of SA-AEF. As shown in Fig. 4, SA-AEF obtains higher accuracy than Concat Fusion and Attention Fusion in most target regions, demonstrating the effectiveness of the proposed fusion design across regional test sets. Dangtu is used as a representative target-region case study in Table 3 to examine the internal contributions of Prompt, CAG, and AEF under a fixed cross-region setting. In particular, removing AEF reduces Acc. from 73.3% to 71.7% and Weighted-F1 from 73.1% to 71.4%, while removing CAG reduces Macro-F1 from 63.4% to 62.1%. These case-study results demonstrate the contributions of source prompts, CAG, and AEF to robust vowel representation on the Dangtu target region.

For phonetic-level interpretability, Fig. 5 visualizes normalized confusion matrices after mapping fine-grained IPA labels into four vowel-height groups. The diagonal dominance indicates that SA-AEF preserves major height distinctions, while most errors occur between adjacent groups, reflecting fine-grained phonetic ambiguity.

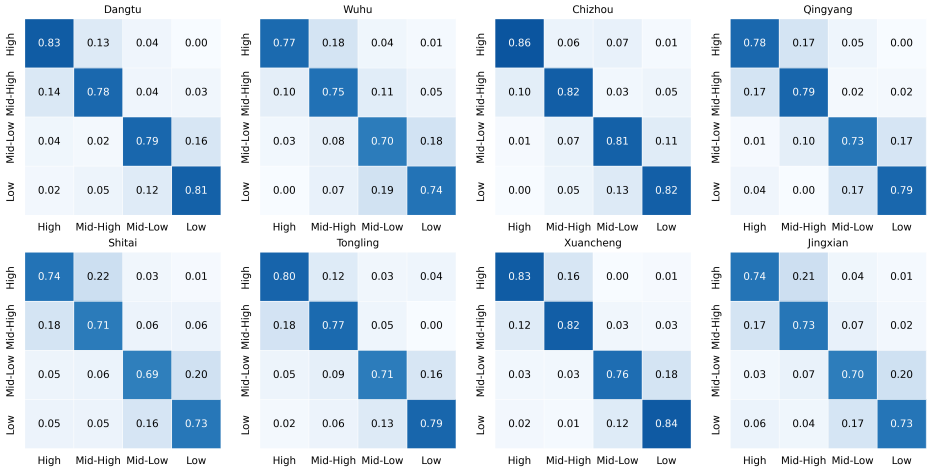


Fig. 5. Phonetic-height-level normalized confusion matrices of SA-AEF on eight representative target regions under cross-region evaluation. Rows and columns denote ground-truth and predicted vowel-height groups, respectively.

5 Conclusion

This paper proposes SA-AEF, a source-aware acoustic enhanced fusion framework for cross-regional Wu dialect vowel recognition. By integrating source-aware tokenization, context-aware recalibration, and cross-source interaction modeling, SA-AEF outperforms the compared baselines on WuVowelSeg. Future work will explore phonetic-knowledge-guided fusion and broader Wu dialect evaluation.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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